

Wi-ViTAL: Domain Generalization of Wireless Human Activity Recognition Using Linear Attention Vision Transformer With Adversarial Learning

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Abstract—The learning-based, passive, device-free wireless human activity recognition (WHAR) systems still face significant challenges, especially in real-world deployments. Environmental differences and domain diversities cause signals collected in the source domain to have a different distribution from those in the target domain, and this affects the accuracy. To achieve domain generalization (DG), a multi-scale linear attention vision transformer (ViT) based feature extractor and domain adversarial learning with Wasserstein distance are proposed. By aligning both marginal and conditional distributions across different source domains, the adversarial learning reduces the differences between trained and unseen domains. As a result, the extracted features become domain-invariant in the latent space, ensuring accuracy is preserved in new or unseen domains. Extensive evaluations using commercial IEEE 802.11ac routers with human activity data collected over different days, environments, human subjects, and obstacle configurations show that the proposed Wi-ViTAL achieves 97.57% average accuracy for five-label classification and more than 76% for eight-label classification in unseen domains. Wi-ViTAL also demonstrates an overall DG improvement compared to other recent benchmarks.

Index Terms—Wireless human activity recognition, domain generalization, multi-scale linear attention, vision transformer, adversarial learning.

I. INTRODUCTION

INTEGRATED Sensing and Communications (ISAC) has become a key feature of 6G wireless [1]. It integrates communication and sensing functions into a unified framework, enabling networks to transmit data while simultaneously performing ubiquitous sensing. This convergence allows for continuous,

real-time monitoring of the environment without interrupting communication processes. The ability to perform non-intrusive and ubiquitous sensing across various domains paves the way for a wide range of advanced applications, including digital twins, smart healthcare, smart homes and offices, safety protection, and so on [2].

Radio frequency (RF) signals encounter scattering, reflection, refraction, and diffraction when passing through transmission paths with targets like static objects or human beings. These interactions alter the properties of the received signals, such as received signal strength indicator (RSSI), channel state information (CSI), Doppler frequency shift and time of flight (ToF), which are utilized to perform RF sensing [3]. There are generally three mainstream approaches to RF sensing, namely millimeter wave (mmWave), impulse-radio ultra-wideband (IR-UWB), and Wi-Fi. The extensive deployment and cost-effectiveness of Wi-Fi make it an ideal candidate for pervasive sensing applications [4].

However, current 802.11 WLAN standards are primarily designed for communication and are not equipped to support Wi-Fi sensing capabilities [5]. To achieve this goal, the IEEE 802.11 working group has established IEEE 802.11bf, which is tasked with defining the essential MAC and PHY procedures to support Wi-Fi sensing in the 2.4 GHz, 5 GHz, 6 GHz, and mmWave bands, including 24 GHz and 60 GHz [6]. Although IEEE 802.11bf provides a unified framework for sensing procedures and modes to measure the channel states and transmission results between the transmitter and receiver, the subsequent processes such as data preprocessing, model design and learning techniques are beyond the scope of 802.11bf. These areas remain open research topics and provide researchers with opportunities to explore advanced algorithms for feature extraction, develop machine learning models to enhance sensing accuracy, and create solutions for cross-domain adaptability [7].

Although Wi-Fi sensing, particularly in the area of wireless human activity recognition (WHAR), has gained significant traction over the past decade, developing models capable of recognizing behavioural patterns across different environments remains a major challenge. Domain differences cause distribution shifts, leading to incorrect recognition; therefore, developing

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a model that can generalize to out-of-distribution (OOD) data is crucial for improving the robustness of Wi-Fi sensing [8]. Domain generalization (DG) addresses OOD-related issues by facilitating the learning of domain-invariant features and apply the knowledge gained from a variety of source domains to previously unseen domains [9]. DG does not require additional data from target domains to fine-tune the pretrained model. This makes DG more practical, as labeled target data is not always available, and expecting users to fine-tune models whenever environments change is neither user-friendly nor cost-effective.

This paper presents an approach that allows the model to learn domain-invariant behavioural patterns from a limited number of source domains, without the need for target-domain data during the training process. A vision transformer (ViT) based feature extractor and domain adversarial learning, leveraging the Wasserstein distance [10] combined with marginal distribution alignment (MDA) and conditional distribution alignment (CDA), are employed to extract domain-invariant features. MDA aims to align domain-specific features across different source domains by minimizing marginal distribution discrepancies, while CDA aligns feature distributions within corresponding labels. With a min-max mechanism between the feature extractor and the domain classifier via the proposed adversarial loss, the feature extractor is prompted to focus on behavioural patterns while simultaneously reducing the distribution discrepancy between source and target domains, projecting features into a shared latent space. The results presented in this paper demonstrate that the proposed DG system, referred to as Wi-ViTAL in the subsequent sections, effectively enables the model to learn domain-invariant features. The key contributions of this work are outlined as follows:

- Wi-ViTAL, a DG model for WHAR, which integrates a ViT-based feature extractor with multi-scale ReLU attention and adversarial learning, is proposed. It requires only source-domain data for training and achieves strong DG performance on unseen target domains, making it practical for real-world applications.
- A novel Wasserstein distance-based adversarial learning approach, combined with MDA and CDA, is introduced to align feature distributions across multiple source domains, both globally and with respect to different activity labels. This allows the feature extractor not only to capture behavioural information but also to map these patterns into a shared latent space with minimal domain influence.
- Wi-ViTAL balances computational efficiency with high classification accuracy, achieving a competitive inference time of approximately 2.1 ms per sample. It also demonstrates the highest average classification accuracy on both five-label (97.57%) and eight-label (76.17%) scenarios when compared to four other state-of-the-art methods.
- An extensive evaluation was conducted on a public dataset collected with commercial IEEE 802.11ac routers rather than older IEEE 802.11n devices, covering five to eight human activities across different days, environments, human subjects and obstacle configurations. The results in this paper provide valuable insights for future deployments of passive, device-free wireless sensing systems, particularly for WHAR applications.

II. RELATED WORKS

Domain adaptation (DA) and domain generalization (DG) are two prominent approaches for addressing domain shift between source and target domains, aiming to make models transferable to new environments [11]. DA assumes the target domain's data is available for fine-tuning the source-trained model. DG, on the other hand, focuses on generalizing a model trained on source-domain data to unseen domains without access to target-domain data during training.

A. DA-Based Approaches

DA, as a branch of transfer learning, aims to improve the performance of the model in the target domains by aligning the feature distributions, which reduces the domain discrepancy and enables the features learned in the source domain to be applied in the target domain [12].

The most widely used DA-based network is domain-adversarial neural network (DANN) [13]. In DANN, a min-max game is played between the feature extractor and the domain classifier. During the training process, the feature extractor is fed data from labeled source and unlabeled target domains, aiming to learn domain-invariant characteristics by fooling the domain classifier [14]. The EI framework [15] leveraged a DANN that eliminates environmental and subject information. It proposed six different loss functions to enhance the generalization ability. In [16] and [17], unsupervised adversarial domain adaptation was employed to train a target encoder to confuse the discriminator and map the data into a domain-invariant feature space. This approach enables the target encoder and the shared classifier to accurately recognize activities in the target domains. [18] and [19] utilized the generative adversarial network (GAN) framework to generate data with features similar to those of the source domain, while deceiving the classifier that distinguishes between the generated data and unlabeled target data to achieve cross-subject transfer learning. Adversarial learning with maximum mean discrepancy (MMD) [20] regularization was employed in both [21] and [22] to align the data distribution over source and target domains. WiCAR [23] expanded the single discriminator into a set of domain discriminators, each focusing on a specific type of domain-specific characteristic, which prevents the mixing of discriminative structures. DGDATA [24] used the gradient reversal layer (GRL) and recognized temporal dependency relations embedded within time series data to improve the classification performance in cross-user HAR. In [25], cross-subject transfer is achieved using combination of CNN, Bi-LSTM, and attention layers, with fine-tuning on target-domain data. Although above DA-based networks can extract domain-invariant features, their performance relies on the availability of data from target domains during the training process. This poses a challenge, as data from target domains is not always accessible in advance.

B. DG-Based Approaches

DG, unlike DA, does not rely on data from the target domains. Instead, it aims to develop a framework capable of generalizing to unseen domains without knowledge of the target

distribution [34]. Achieving DG is particularly essential, as the number of potential domains in the real world is virtually unlimited due to various environmental factors, such as the presence of objects, people, room structures, layouts, etc. Collecting such data from an innumerable number of domains is not feasible, and fine-tuning a model for each domain is too resource-intensive. Therefore, learning generalized patterns from limited training sets not only reduces the burden of data collection and continuous model updating but also makes the approach more practical for real-world applications [35], [36]. AFEEMatNet [28] extracted a frequency domain feature matrix for behaviour recognition, leveraging a combination of CNN and Bi-LSTM to learn transferable characteristics among source environments. [29] introduced SHARP which proposed a technique for cleaning the phase of CSI to capture the Doppler shift of different human activities. In their work, the Doppler shift reveals the presence of dynamic scatterers while ignoring the effect of environmental static objects. Our earlier work [7] leveraged the auto-correlation function and a lightweight feedforward neural network to achieve DG. However, because these approaches rely heavily on traditional signal-processing techniques, some domain-specific information remains in the features after pre-processing, which degrades DG classification accuracy.

Instead of relying on signal processing functions to extract activity-related features, model-based learning can further enhance DG. [26] learned the multi-view invariant feature representations using convolutional and Siamese adversarial networks. [27] improved DG ability by aggregating domain-specific features into unified feature representations, and leveraging MMD to reduce the distribution divergence. WiSR [30] and WiSGP [31], on the other hand, employed random interpolation and subdomain-guided perturbations, aiming to extract generalized representations from multiple source domains while blurring the impact of different domains. [30] also adopted domain adversarial learning but used negative average log-softmax loss to further learn domain-invariant features. AirFi [37] minimized the adversarial loss by imposing the dependency of the feature code extraction on a prior Laplace distribution and consequently reduced reliance on the training dataset. Furthermore, it reduced the distribution variance among different training environments by decreasing MMD. WiPhase [32] fused temporal and subcarrier correlation features extracted from the reconstructed CSI phase to improve cross-domain generalization. In [33], a novel domain generalization model integrating meta-learning initialization, an adaptive channel grouping attention mechanism, and the GRL was proposed to address distribution shift across domains. However, in [30], [31], [37], DG was investigated only in the context of gesture recognition, which involved limited detection areas and focused primarily on arm-level activities and small-scale activity sensing. Such approaches to small-scale motion detection are insufficient to capture the temporal relationships required for the large-scale activity sensing, which involves complex movements of the torso and coordinated body-level activities [38], [39]. Additionally, the above DG approaches focus only on MDA while neglecting CDA within each activity class across different domains.

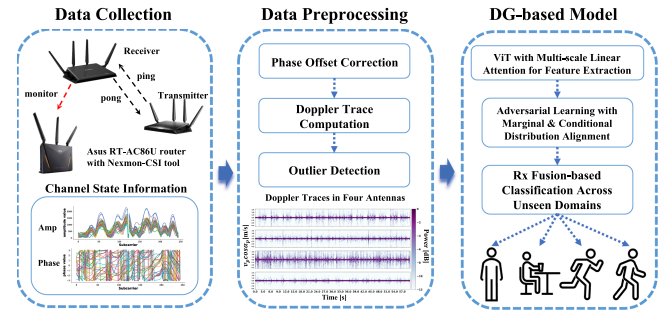


Fig. 1. System framework.

Despite the potential of these DG methods, further exploration is still necessary, particularly for WHAR, due to the inherent complexity and variability of real-world environments. Overall, DG in WHAR research is still in its infancy, and more robust models need to be developed so that they can handle such diverse and ever-changing conditions. Table I compares Wi-ViTAL with state-of-the-art model-based approaches. [18], [19], [21], [22], [24], [27] focus on wearable devices, which, besides being intrusive, capture Inertial Measurement Unit data that is highly correlated with human activities. In contrast, wireless CSI data in WHAR is indirect and less intuitive, often carrying adverse environmental information unrelated to human activities. In other words, CSI not only lacks spatial resolution but also has features that are highly specific to the location and orientation where the activities are performed. Consequently, Wi-ViTAL differs from the prior works in techniques and applications by utilizing a linear attention-based model integrated with the Wasserstein distance to align both marginal and conditional distributions, extracting general patterns from unintuitive CSI to achieve DG.

III. PRELIMINARIES AND PREPROCESSING

Fig. 1 illustrates the overall framework of Wi-ViTAL. A WiFi router, equipped with four antennas and the Nexmon-CSI extraction tool [40], was used to capture CSI transmitted between transceivers. Due to effects of multipath and hardware imperfections, the phase offset in CSI needs to be eliminated. The velocities of various human activities after removing anomalies are recorded as Doppler traces and used as input features for classification. To achieve DG, a ViT-based feature extractor with multi-scale linear attention is employed to capture fine-grained signal behaviours. Adversarial learning enables the feature extractor to acquire domain-invariant subchannel-level features by aligning both marginal and conditional distributions across domains. Finally, the results from the four antennas are fused to make the final decision.

A. Problem Formulation

The general setting of DG involves $\mathbb{N}$ distinct source domains $\mathbb{S}_{\text{train}} = \{\mathbb{S}^d\}_{d=1}^{\mathbb{N}}$ for training, where $\mathbb{S}^d = \{(X_d^{\text{source}}, Y_d^{\text{source}}, d)\}$. Normally, different source domains exhibit varying feature distributions, i.e., $\mathbb{S}^d \neq \mathbb{S}^{d'}$, which means the same activity is represented differently in terms of features across two domains. DG also considers $\mathbb{M}$ unseen target

TABLE I
COMPARISON OF STATE-OF-THE-ART MODEL-BASED SENSING APPROACHES FOR CROSS-DOMAIN HAR

Reference	Attention block	Adversarial block	CDA	Device-less	DA / DG	Cross *	Activity type
Soleimani et al. (2021, 2022) [18], [19]	✗	GAN	✗	✗ (Wearable)	DA	Subject	Daily activities
Suh et al. (2022) [21]	✗	GRL & MMD	✗	✗ (Wearable)	DA	Subject	Daily activities
Suh et al. (2023) [22]	✓	GRL & MMD	✗	✗ (Wearable)	DA	Subject	Daily activities
Ye et al. (2024) [24]	✓	GRL	✗	✗ (Wearable)	DA	Subject	Daily activities
He et al. (2025) [25]	✓	-	✗	✓ (Wireless)	DA	Subject	Gesture / Daily activities
Bai et al. (2020) [26]	✗	Wasserstein	✗	✗ (Wearable)	DG	Subject	Daily activities
Qin et al. (2022) [27]	✗	-	✗	✗ (Wearable)	DG	Subject	Daily activities
Shi et al. (2022) [28]	✗	-	✗	✓ (Wireless)	DG	Env	Daily activities
Meneghello et al. (2023) [29]	✗	-	✗	✓ (Wireless)	DG	Day / Env / Subject	Daily activities
Liu et al. (2024) [30]	✗	GRL	✗	✓ (Wireless)	DG	Env / Loc / Subject	Gesture
Liu et al. (2024) [31]	✗	-	✗	✓ (Wireless)	DG	Env / Loc / Subject	Gesture
Chen et al. (2024) [32]	✓	-	✗	✓ (Wireless)	DG	Env / Subject	Daily activities
Fan et al. (2025) [33]	✓	GRL	✗	✓ (Wireless)	DG	Env / Subject	Daily activities
Ours (Wi-ViTAL)	✓	Wasserstein	✓	✓ (Wireless)	DG	Day / Env / Subject	Daily activities

domains $\mathbb{T}_{\text{test}} = \{\mathbb{T}_d^M\}_{d=1}^M$, with $\mathbb{T}^d = \{(X_d^{\text{target}}, Y_d^{\text{target}})\}$. The goal of DG is to tackle the challenge of learning a robust model that captures domain-invariant patterns from the source domains and generalizes effectively to unseen domains. Given $\mathbb{N}$ source domains, the prediction error on unseen domains needs to be minimized, expressed as $\min \mathbb{E}_{(X,Y) \in \mathbb{T}_{\text{test}}} [\mathcal{L}(f(X), Y)]$, where $\mathbb{E}$ denotes the expectation over the target domain, $\mathcal{L}$ is the loss function, and $f(X)$ outputs a vector of logits corresponding to each class.

B. Phase Offset Correction

OFDM is a widely used modulation scheme in IEEE 802.11 WLAN standards, and encodes a data stream into multiple orthogonal subcarriers, which are transmitted in parallel. CSI is the metric used in OFDM systems, capturing amplitude and phase variations across subcarrier frequencies during signal propagation between transceivers [41]. In practice, CSI is represented as a complex matrix derived through the fast Fourier transform, estimating channel frequency response (CFR) for each subcarrier. The CFR $H_{n,k}$ at the n^{th} packet and k^{th} subcarrier can be expressed as:

$$H_{n,k} = \sum_{p=0}^{P-1} A_{n,p} e^{-j2\pi(f_c + k\Delta f)\tau_{n,p}} = A_{n,k} e^{j\theta_{n,k}} \quad (1)$$

where $A_{n,p}$ and $\tau_{n,p}$ denote the attenuation and time delay for the n^{th} packet along the p^{th} path, respectively. The term $f_c + k\Delta f$ represents the frequency of the k^{th} subcarrier relative to the main carrier frequency f_c , with $k \in \{-\frac{K}{2}, \dots, \frac{K}{2} - 1\}$. Consequently, $A_{n,k}$ and $\theta_{n,k}$ represent the attenuation and phase shift during the transmission of the OFDM symbol.

In practice, the undesired amplitude offset can be effectively mitigated by normalizing it with the mean amplitude of the monitored subchannels within the channel coherence time [42]. Phase information is influenced by the propagation environment and hardware imperfections, resulting in a phase offset $\Delta\theta$. More specifically, phase offset is caused by channel frequency offset (CFO), the phase-locked loop phase offset (PPO) and the phase

ambiguity (PA) [43], [44]. Sampling frequency offset (SFO) and packet detection delay (PDD) alter the perceived arrival time of the signal at the receiver [45]. Therefore, the equation for CFR with phase offset is given by (2).

$$\tilde{H}_{n,k} = A_{n,k} e^{j(\theta_{n,k} + \Delta\theta_k)}, \text{ with } \Delta\theta_k = -2\pi k \frac{\Delta\tau}{T} + \Delta\phi \quad (2)$$

where the overall phase shift for the k^{th} subcarrier is $\Delta\theta_k$. The phase offset caused by CFO, PPO and PA is denoted as $\Delta\phi$, while the time delay resulting from SFO and PDD is represented as $\Delta\tau$.

The expression $\mathcal{X}_{n,k}$ reformulates (2) into a matrix form, capturing the influence of the time delay and phase shift as follows:

$$\mathcal{X}_{n,k} = e^{j\Delta\phi} \begin{bmatrix} e^{-j2\pi k(\Delta\tau + \tau_{n,0})/T} \\ \vdots \\ e^{-j2\pi k(\Delta\tau + \tau_{n,P'-1})/T} \end{bmatrix} \circ \begin{bmatrix} A_{n,0} e^{-j2\pi f_c \tau_{n,0}} \\ \vdots \\ A_{n,P'-1} e^{-j2\pi f_c \tau_{n,P'-1}} \end{bmatrix} \quad (3)$$

$$r = e^{j\Delta\phi} [A_{n,0} e^{-j2\pi f_c \tau_{n,0}} \dots A_{n,P'-1} e^{-j2\pi f_c \tau_{n,P'-1}}]^T \quad (4)$$

where the actual paths p' , ranging from 0 to $P' - 1$, may differ from the theoretical paths p . The left matrix in (3), determined by the time delay, is assumed for different paths based on the reasonable frequency range [29], while r is calculated by solving the minimization problem in [46] and detailed in the Appendix.

The following phase offset correction procedure is based on [29] and only the key aspects are emphasized here. Let p^* denote the reference path with the highest amplitude of r , assumed to correspond to a static component. The phase offset in $\mathcal{X}_{n,k}$ is eliminated by applying the complex conjugate of $\mathcal{X}_{n,k}$ for the path p^* . Therefore, the estimated CFR without phase

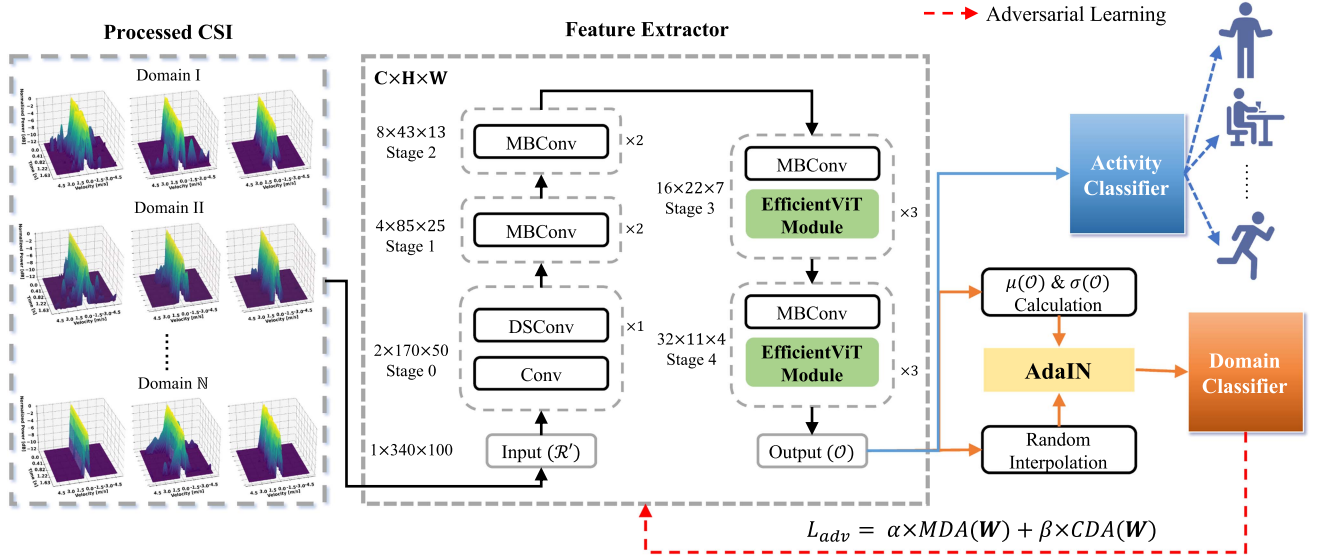


Fig. 2. Wi-ViTAL, a DG-based model architecture using high computation-efficient convolutions and EfficientViT modules with Wasserstein distance-based adversarial learning under the marginal and conditional distribution alignments.

offset, summed over all paths, can be expressed as:

$$\begin{aligned}\hat{H}_{n,k} &= A_{n,p^*} \sum_{p'=0}^{P'-1} A_{n,p'} e^{-j2\pi(f_c+k\Delta f)(\tau_{n,p'}-\tau_{n,p^*})} \\ &\approx A_{n,p^*} e^{j2\pi(f_c+k\Delta f)\tau_{n,p^*}} H_{n,k}\end{aligned}\quad (5)$$

FFT is applied to $\hat{H}_{n,k}$ to capture frequency shifts or the Doppler effect caused by activities. The i^{th} block of $\hat{H}_k$, from $\hat{H}_{i,k}$ to $\hat{H}_{i+W_S-1,k}$ with length W_S , is expanded to N_S to enhance the granularity for activity recognition. Therefore, the Fourier transform is given as follows:

$$\mathcal{F}_k[m] = \sum_{n=0}^{N_S-1} \hat{H}_k[n+i] W_k[n] e^{-j\frac{2\pi}{N_S} mn} \quad (6)$$

where $m \in [-N_S/2, N_S/2 - 1]$ represents the Doppler index, and $W_k[n]$ is the Hanning function, which effectively reduces spectrum leakage and provides a smooth transition at the boundaries of the signal.

C. Doppler Trace Estimation and Outlier Detection

To obtain Doppler traces, calculating the time delay for packet n along path p is necessary. This delay depends on the initial position and the velocity of scatterers and is represented as:

$$\tau_{n,p} = \frac{\mathcal{L}_p + \Delta\mathcal{L}_{n,p}}{c}, \text{ with } \Delta\mathcal{L}_{n,p} = -v_p \cos(\alpha_p) n T_c \quad (7)$$

where $\mathcal{L}_p$ is the distance between the initial position of the scatterer and the receiver, $\Delta\mathcal{L}_{n,p}$ represents the scatterer's movement, v_p is the velocity of the scatterer, $\cos\alpha_p$ denotes its moving direction, and T_c is the sampling period. Therefore, (6) is reformulated as (8) to emphasize the Doppler traces $v_p \cos\alpha_p$, which

reflect the monitored motions and are environment-invariant.

$$\begin{aligned}\mathcal{F}_k[m] &\approx \sum_{n=0}^{N_S-1} A_{n+i,p^*} \sum_{p=0}^{P'-1} (A_{n+i,p} e^{-j2\pi(f_c+k\Delta f)(\frac{\mathcal{L}_{p-p^*}}{c})}) \\ &\times W_k[n] e^{j2\pi(f_c v_p \cos\alpha_p (n+i) T_c / c - mn / N_S)}\end{aligned}\quad (8)$$

To ensure data quality, outliers from the four antennas need to be removed. A majority vote criterion is employed such that if the same packet is identified as an outlier by more than two antennas, it is subsequently eliminated. The interquartile range (IQR) is applied to remove outliers within each time window of length W_S in the magnitude matrix $\mathcal{R}$, where $\mathcal{R}$ is obtained by summing the product of $\mathcal{F}_k[m]$ and its complex conjugate across subcarriers. The resulting outlier-free matrix, denoted as $\mathcal{R}'$, is subsequently used as input for model training and testing.

IV. MODEL ARCHITECTURE

The proposed Wi-ViTAL model architecture, designed to achieve DG and illustrated in Fig. 2, comprises four essential components, namely the feature extractor, activity classifier, domain classifier, and adversarial learning module. The processed CSI data collected in multiple source domains are used to train and evaluate the model. The feature extractor is designed to capture activity-related features using enhanced multi-scale ReLU linear attention and highly efficient convolutional blocks. The extracted features contain domain-invariant information, reflecting how activities induce changes in CSI within the shared latent space. The output $\mathcal{O}$ from the extractor is then fed into the activity classifier, which is responsible for identifying different activities. In parallel, domain differences result in distinct variations across subchannels, reflected in both $\mu(\mathcal{O})$ and $\sigma(\mathcal{O})$. Consequently, adaptive instance normalization (AdaIn) [47] transforms the interpolated features to domain-specific features, attempting to blur activity-related information.

Finally, the domain classifier performs the domain recognition. To ensure the model's robustness, the adversarial learning module is implemented to create a competitive mechanism between the feature extractor and the domain classifier. The adversarial module attempts to inhibit the domain classifier's ability to detect the domain-specific information, thereby compelling the feature extractor to focus on the activity-related information while minimizing the specific environmental effects.

A. Feature Extractor With Vision Transformer

The feature extractor adopted in Wi-ViTAL was initially introduced by [48], which adopts a lightweight ViT model for high-resolution dense prediction. High-resolution dense prediction focuses on capturing detailed information within images or videos. Similarly, CSI captures variations across multiple subcarriers, reflecting distinct patterns in both signal amplitude and phase information. By concentrating on the detailed variations among the subcarriers, the ViT model distinguishes between similar yet different motions.

The backbone of the feature extractor consists of five stages, where the spatial dimensions of features gradually decrease while the number of channels increases. The EfficientViT modules introduced in [48] are integrated in stages 3 and 4. Stage 0 comprises a standard convolutional layer followed by depthwise separable convolutions (DSConv). The standard convolution layer acts as the initial feature extractor, capturing fundamental elements such as variations in signal strength, phase shifts, and simple patterns. Subsequently, the DSConv, combining depthwise convolution (DWConv) and pointwise convolution, is utilized for further processing. The DWConv emphasizes spatial information, while the pointwise convolution learns correlations across channels. This design captures both spatial and channel information, simultaneously reducing the computational complexity and making it suitable for the deployment on devices with lower computational power. The mobile inverted bottleneck convolution (MBConv) is also a key building block in the backbone. MBConv can capture more complex features by increasing the number of channels through a 1×1 convolution. Then, DSConv, one element in MBConv, works for downsampling with stride 2. Consequently, MBConv maintains efficiency while enhancing the capacity to capture abstract patterns and relationships.

The EfficientViT modules leverage ReLU linear attention [49] to achieve both the global receptive field and linear computational complexity. Let $x \in \mathbb{R}^{N \times F}$ denote N feature vectors, each with F dimensions, corresponding to a sample of processed CSI signals $\mathcal{R}'$. The generalized form of softmax attention $\mathcal{O}_i$ for any similarity function is defined as:

$$\mathcal{O}_i = \sum_{j=1}^N \frac{\text{Sim}(Q_i, K_j)}{\sum_{j=1}^N \text{Sim}(Q_i, K_j)} V_j \quad (9)$$

where i and j represent the i^{th} row and j^{th} row of the matrix, $Q = xW_Q$, $K = xW_K$, $V = xW_V$, $W_Q/W_K/W_V \in \mathbb{R}^{F \times d}$, and W_Q , W_K , and W_V are learnable projection matrices, respectively. The Q , K , V are referred to as the "queries", "keys" and "values". Therefore, the similarity function for ReLU linear

attention is $\text{Sim}(Q, K) = \text{ReLU}(Q)\text{ReLU}(K)^T$, and $\mathcal{O}_i$ can be simplified as:

$$\begin{aligned} \mathcal{O}_i &= \sum_{j=1}^N \frac{\text{ReLU}(Q_i)\text{ReLU}(K_j)^T}{\sum_{j=1}^N \text{ReLU}(Q_i)\text{ReLU}(K_j)^T} V_j \\ &= \frac{\text{ReLU}(Q_i)(\sum_{j=1}^N \text{ReLU}(K_j)^T V_j)}{\text{ReLU}(Q_i)(\sum_{j=1}^N \text{ReLU}(K_j)^T)} \end{aligned} \quad (10)$$

This results in linear computational cost, offering greater efficiency and usability compared to softmax attention because $\sum_{j=1}^N \text{ReLU}(K_j)^T V_j$ and $\sum_{j=1}^N \text{ReLU}(K_j)^T$ are computed once and reused for each query. Since $\text{ReLU}(x) = \max(0, x)$ lacks a non-linear similarity function and a powerful concentrated mechanism, thus the feedforward neural network with DWConv is employed to enhance local information extraction. Also, $Q/K/V$ tokens are extended to multi-scale tokens via lightweight small-kernel DWConv and GConv. This multi-scale ReLU linear attention captures features at different scales while maintaining the computational efficiency. This mitigates the weakness of ReLU linear attention in local information extraction and enables the global receptive field.

B. Activity and Domain Classifiers

Given the feature extractor's strong ability to capture contextual information, the activity classifier is designed to recognize activities with minimal computational cost. A 1×1 convolution is employed to integrate information across channels, followed by average pooling to reduce spatial dimensions while preserving global feature representations. Consequently, the fully connected network maps the input features to the final class predictions.

The domain classifier is also built with the same structure as the activity classifier. To enhance domain classification, an activity randomization block [30] is placed before the domain classifier. This block applies random interpolation of activity-specific features to bias the representation toward domain information, making the domain-related components more prominent. It also estimates the variation across subcarriers, characterized by the mean $\mu(\mathcal{O})$ and variance $\sigma(\mathcal{O})$. Subsequently, AdaIN [47], a style-transfer method, is employed to align the randomized features with domain-specific characteristics using $\mu(\mathcal{O})$ and $\sigma(\mathcal{O})$. This process transforms the fuzzy activity features into the subcarrier domain, ensuring that the model captures domain-specific details.

C. Adversarial Learning Between Feature Extractor and Domain Classifier

Adversarial learning is employed to direct the feature extractor toward activity-specific information while reducing the influence of domain differences. It weakens the domain classifier's ability to discern where the motions are performed, leading to lower domain classification accuracy. Besides, the activity classifier maintains high accuracy in recognizing human activities even in unseen domains.

Wasserstein distance [50] is a metric that measures the difference between two probability distributions. It represents the minimum cost required to transform one distribution to another. A lower Wasserstein distance between features from different domains indicates that less effort is needed to transfer knowledge from the source domain to the target domain. The equation for Wasserstein distance is given by:

$$\mathbf{W}(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathbf{L}_1} \int_{\mathbf{L}_2} \gamma(l_1, l_2) \|l_1, l_2\|_2 d(l_1) d(l_2)$$

$$\text{with } \int \gamma(l_1, l_2) d(l_2) = \mu(l_1), \int \gamma(l_1, l_2) d(l_1) = \nu(l_2) \quad (11)$$

where $\mu(l_1)$ and $\nu(l_2)$ represent the probability distributions of the random variables l_1 and l_2 in the spaces $\mathbf{L}_1$ and $\mathbf{L}_2$, respectively. The joint distribution $\gamma(l_1, l_2)$ is over the combined space $\mathbf{L}_1 \times \mathbf{L}_2$, with $\Pi(\mu, \nu)$ denoting the set of all possible joint distributions between μ and ν . In this context, Euclidean distance $\|l_1, l_2\|_2$ quantifies the degree of feature discrepancy between domains, while $\gamma(l_1, l_2) \|l_1, l_2\|_2$ represents the total transport cost. Besides, $\inf$ corresponds to the minimum transport cost among all feasible transport possibilities. Additionally, incorporating the entropy term $\xi S(\gamma)$ in (11) offers an alternative approach to achieve a smoother transport plan and faster convergence speed, where ξ is a regularization parameter, and i.e., $S(\gamma)$ is the Shannon entropy, given by $\int_{\mathbf{L}_1} \int_{\mathbf{L}_2} \gamma(l_1, l_2) \log(\gamma(l_1, l_2)) d(l_1) d(l_2)$.

Since the domain-related feature differences are minimized using Wasserstein distance, the domain classifier finds it more difficult to distinguish between different domains; therefore, an adversarial relationship is established between the domain classifier and the feature extractor. This mechanism prompts the feature extractor to focus on generating domain-invariant features. Additionally, MDA and CDA [51] are integrated into the adversarial learning process. Instead of only aligning distributions between source and target domains in terms of marginal distribution, MDA and CDA are both integrated with the Wasserstein distance, aiming to minimize the feature discrepancies both globally and with respect to different labels. The adversarial loss related to Wasserstein distance with MDA and CDA is represented as:

$$\begin{aligned} \mathcal{L}_{adv}[\mathbf{T}(\mathbf{X}), \mathbf{T}(\mathbf{X})|c] &= \sum_{i=1}^{\mathbb{N}} \sum_{j=i+1}^{\mathbb{N}} [\alpha \mathbf{W}(\mathbf{T}(\mathbf{X}_i), \mathbf{T}(\mathbf{X}_j)) \\ &+ \beta \sum_{c=1}^{\mathbb{C}} \mathbf{W}([\mathbf{T}(\mathbf{X}_i)|c_i = c], [\mathbf{T}(\mathbf{X}_j)|c_j = c])] \quad (12) \end{aligned}$$

where $\mathbb{N}$ and $\mathbb{C}$ represent the maximum number of source domains and classes that can be selected during the training process, respectively. The Wasserstein distance $\mathbf{W}(\mathbf{T}(\mathbf{X}_i), \mathbf{T}(\mathbf{X}_j))$ measures the alignment between the marginal distributions of domain $\mathbf{X}_i$ and $\mathbf{X}_j$. Specifically, the representation $\mathbf{T}$ refers to the log-softmax transformation applied to the outputs of domain classifier. For CDA, the loss function is also to align the domain distributions by considering the activity labels. α and β are hyperparameters that control the adversarial degree during model training.

Algorithm 1: The Pseudocode of the Model Training.

Input: The multiple source data $\mathbb{S}$, batch size $\mathcal{S}$, number of epochs M , learning rate η , the feature extractor $\mathbf{G}_\theta$, the activity classifier $\mathbf{C}_\varphi$, the domain classifier $\mathbf{D}_\psi$

Result: Feature extractor and activity classifier parameters θ and φ

```

1 Initialize  $\mathbf{G}_\theta, \mathbf{C}_\varphi, \mathbf{D}_\psi$ ;
2 for  $epoch = 0$  to  $M - 1$  do
3   Sample a data batch  $(X_d^{\text{source}}, Y_d^{\text{source}}, d)$  with size  $\mathcal{S}$ ;
4   Activity and domain recognition:
5      $\mathcal{O} = \mathbf{G}_\theta(X_d^{\text{source}})$ ;
6      $\hat{Y} = \mathbf{C}_\varphi(\mathcal{O})$ ;
7      $\mathcal{L}_\varphi = \text{CrossEntropy}(\hat{Y}, Y_d^{\text{source}})$ ;
8     Obtain the domain-related components, the mean  $\mu(\mathcal{O})$  and variance  $\sigma(\mathcal{O})$ ;
9      $\mathcal{O}' = \text{SHUFFLE}(\mathcal{O})$ ;
10     $\lambda = \text{Random}(0, 1)$ ;
11     $\tilde{\mathcal{O}} = \lambda \cdot \mathcal{O} + (1 - \lambda) \mathcal{O}'$ ;
12     $\mathcal{O}_d = \sigma(\mathcal{O}) \left( \frac{\tilde{\mathcal{O}} - \mu(\tilde{\mathcal{O}})}{\sigma(\tilde{\mathcal{O}})} \right) + \mu(\mathcal{O})$ ;
13     $\hat{d} = \mathbf{D}_\psi(\mathcal{O}_d)$ ;
14     $\mathcal{L}_\psi = \text{CrossEntropy}(\hat{d}, d)$ ;
15     $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_\varphi + \mathcal{L}_\psi)$ ;
16     $\varphi \leftarrow \varphi - \eta \nabla_\varphi (\mathcal{L}_\varphi + \mathcal{L}_\psi)$ ;
17     $\psi \leftarrow \psi - \eta \nabla_\psi (\mathcal{L}_\varphi + \mathcal{L}_\psi)$ ;
18  Adversarial learning between  $\mathbf{G}_\theta$  and  $\mathbf{D}_\psi$ :
19    if  $epoch \% 2 == 0$  then
20       $\mathbf{T}(\hat{d}) = -\log\_softmax(\hat{d})$ ;
21       $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{adv}[\mathbf{T}(\hat{d}), \mathbf{T}(\hat{d}|Y_d^{\text{source}})]$ 
22    end
23 end
```

Moreover, the feature extractor $\mathbf{G}_\theta$, activity classifier $\mathbf{C}_\varphi$, and domain classifier $\mathbf{D}_\psi$ are continuously updated to achieve optimal performance. The adversarial learning is applied every two epochs, rather than at every step, to maintain activity recognition performance while reducing domain classification accuracy. This controlled execution prevents both classifiers from degrading simultaneously and addresses the generalization flaw that averages accuracy across domains. As a result, a domain-invariant representation is created, which reduces the impact of multipath effects and ultimately improves domain generalization. The pseudocode for model training is provided in Algorithm 1.

V. EXPERIMENTS

A. Dataset Introduction

The performance of Wi-ViTAL was evaluated on a WHAR dataset (2023) [57], which featured notable domain diversity across different days, environments, participants and obstacle configurations. The testbed in this dataset comprised two single-antenna Netgear X4S AC2600 IEEE 802.11ac routers, which served as the transmitter and receiver, communicating over a single spatial stream at a rate of 173 packets per second. The bi-static sensing network operated on channel 42, over an

TABLE II
MEASUREMENT CONDITIONS OF IEEE 802.11ac DATASET

Set	Device pos.	Environment	Campaigns	Obst.	Person, Pi	W × L × H (m)	Furniture
AR-1	M1-Tx-Rx	bedroom	a-b-c-d-e	✗	P1	$5 \times 6 \times 4$	bookcase, 2 beds, desk, chairs
AR-2	M1-Tx-Rx	bedroom	a	✗	P2		
AR-3	M2-Tx-Rx	bedroom	a-b	✓	P1		
AR-4	M2-Tx-Rx	bedroom	a	✓	P2		
AR-5	M3-Tx1-Rx1	living room	a-b	✗	P1	$5 \times 6 \times 4$	armchair, TV cabinet, table, chairs, 2 sofas
AR-6	M3-Tx1-Rx1	kitchen	a	✗	P1	$3.5 \times 3 \times 3.2$	hob, 2 cabinets
AR-7	M3-Tx1-Rx1	laboratory	a	✗	P3	$7.5 \times 3.5 \times 2.9$	4 desks, 6 workstations, 6 displays

TABLE III
PUBLIC DATASET COMPARISON FOR WI-FI-BASED WHAR

Dataset	No. of days per environment	No. of environments	No. of involved subjects	No. of activities	NLOS	WLAN standard	Bandwidth
[52]	1	1	6	6	✗	802.11n	20 MHz
[53]	1	3	10	16	✗	802.11n	20 MHz
[54]	1	3	1	7	✗	802.11n	40 MHz
[55]	1	4	1	11	✗	802.11ac	80 MHz
[56]	1	3	1	4	✗	802.11ac	80MHz
[40]	1	3	1	5 ~ 8	✗	802.11ac	40 ~ 80 MHz
This paper [57]	1 ~ 5	5	3	5 ~ 8	✓	802.11ac	80 MHz

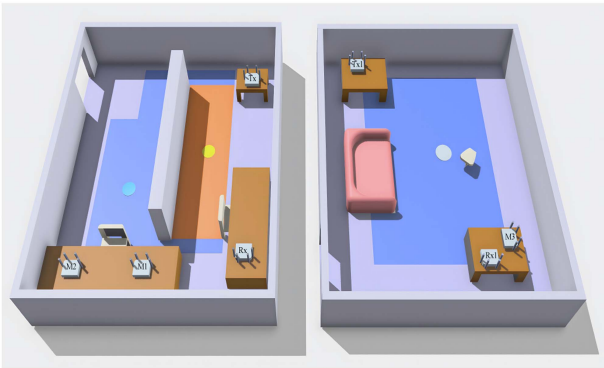


Fig. 3. Experimental setups (left: bedroom with a wood bookcase in the middle; right: layout of key components in the living room, kitchen, and laboratory).

80 MHz bandwidth with a center frequency of 5210 MHz. In addition, an Asus RT-AC86 U router was used to capture CSI packets, which was equipped with four antennas and the Nexmon CSI tool. This study focuses on sets from AR-1 to AR-7, which are designed for human activity sensing. These sets contain five activities: empty (E), sitting still (L), walking (W), running (R), and jumping (J). Some sets also include additional human activities, including sitting down/standing up (C), standing still (S), and arm exercises (H). Table II summarizes the measurement conditions. The colors correspond to the monitored areas shown in Fig. 3. The orange and blue zones indicate the areas where the volunteers performed walking and running experiments, respectively. In the bedroom, the direct propagation path is obstructed by a bookcase in the middle when the monitor is at M2, creating a non-line-of-sight (NLOS) scenario. The sitting still and sitting down/standing up activities took place at the locations of chairs. Other activities, including standing still,

jumping, and arm exercises, were conducted at the colored circles. Each volunteer repeatedly performed the task during the trace acquisition period. Notably, Fig. 3 illustrates two typical layouts. While the specific furniture arrangements vary across the different environments depicted in Table II, the positions of transceivers and CSI monitoring device remain unchanged. This dataset has been compared with other human activity sensing datasets in [57], as shown in Table III, which confirms that it offers richer environmental variability and greater time diversity, along with more detailed CSI information. Furthermore, the CSI collected using IEEE 802.11ac consists of 256 subcarriers, offering a higher sensing resolution than most existing datasets, which are based on the older IEEE 802.11n standard with only up to 128 subcarriers.

B. Hyperparameter Setup

The preprocessing hyperparameters are set the same as those in [29] to ensure a fair performance comparison. Out of the 256 subcarriers, 242 are valid data subcarriers with indices ranging from -122 to -2 and from 2 to 122. To calculate Doppler traces, a window of $W_S = 31$ consecutive sanitized CSI packets are used and expanded to $N_S = 100$ via zero-padding during the FFT, which enhances the frequency resolution. W_S also serves as the range for outlier detection in Section III-C, based on the assumption that speed is generally continuous and abrupt changes are rare over short time intervals. The preprocessing results (e.g., for sitting down/standing up and running) are presented in Fig. 4. During the training phase, Doppler traces comprising 340 Doppler vectors (approximately 2 seconds) are fed to the Wi-ViTAL model for human activity recognition, providing a practical window size for real-time recognition. α and β for MDA and CDA during adversarial learning are

TABLE IV
PERFORMANCE EVALUATION ACROSS DIFFERENT DOMAINS

Set		AR-1a	AR-1b	AR-1c	AR-1d	AR-1e	AR-2a	AR-3a	AR-3b	AR-4a	AR-5a	AR-5b	AR-6a	AR-7a
Data type		train-test	train-test	train-test	test	test	test	test	test	test	test	test	test	test
Empty	Precision	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9843	1.0000	1.0000	1.0000	1.0000
	Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.9599	0.9952	1.0000	1.0000	1.0000	0.9866	1.0000	1.0000
Sitting	Precision	1.0000	1.0000	1.0000	1.0000	1.0000	0.9600	0.9951	1.0000	1.0000	1.0000	0.9875	1.0000	1.0000
	Recall	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9841	1.0000	0.9895	1.0000	0.9645
Walking	Precision	1.0000	1.0000	1.0000	0.9601	1.0000	0.9980	1.0000	0.9223	0.9898	0.9927	1.0000	0.6455	1.0000
	Recall	1.0000	1.0000	1.0000	1.0000	0.9982	1.0000	0.9978	0.9958	0.7679	1.0000	1.0000	0.5888	0.9633
Running	Precision	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9978	0.9870	0.8003	0.9981	1.0000	0.5618	0.9603
	Recall	1.0000	1.0000	1.0000	0.9557	1.0000	0.9979	1.0000	0.9192	0.9936	0.9924	1.0000	0.9307	1.0000
Jumping	Precision	1.0000	1.0000	1.0000	1.0000	0.9979	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9509	0.9578
	Recall	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9922	0.9980	0.9979	1.0000	0.2162	1.0000
Mean	Accuracy	1.0000	1.0000	1.0000	0.9953	0.9996	0.9900	0.9985	0.9831	0.9521	0.9982	0.9971	0.7855	0.9851

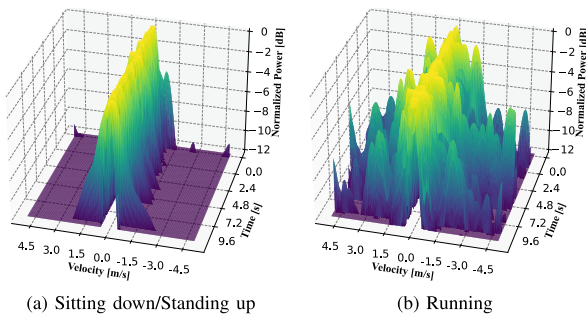


Fig. 4. Doppler traces expressed by velocity ($v_p \cos \alpha_p$) over a 10-second period for two example Activities.

empirically set to $5e^{-3}$ and $1e^{-2}$, respectively, thereby balancing behaviour classification accuracy and the potential reduction caused by adversarial learning. The model is trained across three different domain settings, including temporal, environmental, and subject variations. The three input sets are split into 60% for training, 20% for validation, and 20% for testing. The data from the remaining unseen target domains are used exclusively for testing. Since Wi-ViTAL converges quickly with a small number of labels, the number of training epochs is set to 10 for five-label classification, while 100 epochs for eight-label classification with an early stopping criterion of 10 epochs. For comparison, other models [27], [29], [30], [33] are trained for 100 epochs with a 10-epoch early stopping criterion to ensure that the models are well-trained for both the five-label and eight-label settings. The batch size is fixed at 32 for both training and testing. In this work, PyTorch is used on a workstation equipped with an NVIDIA RTX 3090 GPU, an Intel i9-12900K CPU, and 64 GB of RAM.

C. Performance Assessment for Five Classes

To ensure consistency with [29], five of the most representative activities, namely empty, sitting, walking, running, and jumping, are evaluated across multiple domains. Three source domains (AR-1a, AR-1b, AR-1c) with temporal variation are used for model training. The best model is selected based on validation performance and is subsequently tested on unseen domains, ranging from AR-1 d to AR-7a. Overall, the generalization performance is satisfactory across most target domains,

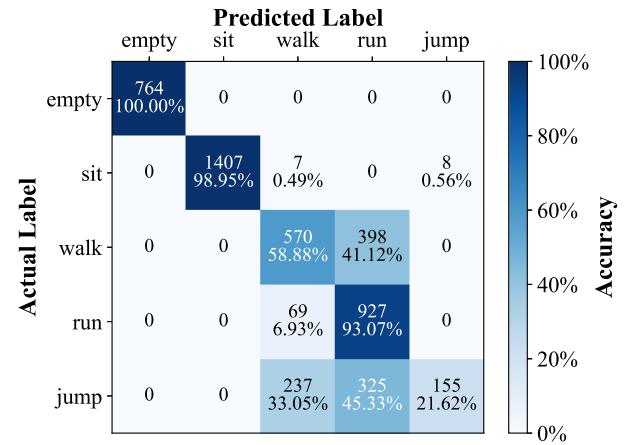


Fig. 5. Confusion matrix for AR-6a (kitchen).

with precision and recall scores consistently close to 1.00, as shown in Table IV.

However, a significant drop in performance is observed in the kitchen (AR-6a), especially for activities such as walking, running, and jumping. Walking and running are inherently prone to misclassification due to their similar movement patterns. Despite this, the model achieves a relatively high recall of 0.9307 for running, indicating that running can generally be recognized correctly, although some instances of high-velocity walking are confused with running. A significant portion of jumping instances is also misclassified as either walking or running, as shown in the confusion matrix in Fig. 5. These misclassifications are likely attributed to the preprocessing step, which primarily focuses on movement speed. In addition, inconsistencies in how test subjects perform the same activity at different times and locations also lead to errors. Another factor contributing to these misclassifications is the disparity between the training and testing environments. For instance, the kitchen differs significantly from the bedroom in terms of size and furniture arrangement, leading to variations in multipath effects and signal attenuation, which consequently impact the classification accuracy. Yet, the model still achieves excellent performance in the bedroom with NLOS conditions, as well as in the living room and laboratory, demonstrating its strong generalization capability.

TABLE V
DG PERFORMANCE COMPARISON FOR FIVE-LABEL CLASSIFICATION

Methods	AR-1a	AR-1b	AR-1c	AR-1d	AR-1e	AR-2a	AR-3a	AR-3b	AR-4a	AR-5a	AR-5b	AR-6a	AR-7a	Mean
AFFAR	1.0000	1.0000	1.0000	1.0000	1.0000	0.9985	0.9996	0.9845	0.7578	0.9173	0.9392	0.7489	0.9586	0.9465
SHARP	1.0000	1.0000	0.9733	0.9880	1.0000	0.9753	0.9973	0.9603	<u>0.9265</u>	0.9789	<u>0.9960</u>	0.6304	<u>0.9722</u>	<u>0.9537</u>
WiSR	0.9809	1.0000	0.9569	0.9809	0.9833	0.9247	0.9791	0.9460	0.8097	0.9741	0.7950	0.7138	0.9304	0.9211
Meta-EMCAN	1.0000	1.0000	1.0000	<u>0.9980</u>	1.0000	0.9985	0.9961	0.9086	0.7688	<u>0.9942</u>	0.9928	0.6501	0.8515	0.9353
Wi-ViTAL	1.0000	1.0000	1.0000	0.9953	<u>0.9996</u>	0.9900	0.9985	<u>0.9831</u>	0.9521	0.9982	0.9971	0.7855	0.9851	0.9757

D. DG Performance Evaluation

Four approaches, namely AFFAR (2022) [27], SHARP (2023) [29], WiSR (2024) [30], and Meta-EMCAN (2025) [33], are compared with Wi-ViTAL, as they have demonstrated superior DG performance. Table V demonstrates the DG performance across various domains ranging from AR-1a to AR-7a. The data used for training, validation, and testing are the same as those in Table IV. As shown, Wi-ViTAL outperforms the other four approaches in most domains, particularly in unseen environments that differ significantly from the source domains, such as NLOS scenarios, living room, and laboratory. AFFAR and Meta-EMCAN exhibit slightly better classification performance than Wi-ViTAL in certain bedroom environments, although the maximum difference is less than 0.85%. Table V also shows that Wi-ViTAL consistently maintains the second-best performance in the bedroom, except in the AR-1 d scenario, with only a 0.27% gap. Wi-ViTAL's overall generalization capability, as demonstrated by an average accuracy of 97.57%, is higher than that of the other four models. Notably, the other models struggle in AR-4a, which features NLOS conditions and a change in the volunteer from P1 to P2. Nevertheless, Wi-ViTAL still achieves more than 95% accuracy, outperforming the other methods, which score 75.78%, 92.65%, 80.97%, and 76.88%, respectively. In addition, the proposed model achieves the highest classification accuracy from AR-4a to AR-7a, despite substantial changes in environmental settings and participants.

Nevertheless, all models, including Wi-ViTAL, experience a significant performance drop in AR-6a (kitchen). This is because the materials and arrangement of furniture in the kitchen differ greatly from those in bedrooms. Kitchens typically contain large metal appliances, ceramic tiles, and glass surfaces, all of which are packed into a relatively small space. Such a high density of reflective materials leads to stronger multipath interference, altering the data distribution for the same activity. To further investigate this, the Kolmogorov-Smirnov (KS) test [58] is employed to measure the distribution differences. Raw data with dynamic motions from the bedroom, NLOS scenarios, living room, kitchen, and laboratory, each with dimensions $4 \times 100 \times 4000$, are selected to compare the degree of domain discrepancies. These data represent more than 20 seconds of consecutive activity, collected by four antennas, using the bedroom data as the reference. The asymptotic critical value $D_{0.05}$ at a 5% significance level is 2.15×10^{-3} , based on these dimensions. Fig. 6 shows how much the data distributions from unseen domains differ from that of the bedroom data used in the training set. As expected, the largest discrepancy (the highest KS statistic values) is observed in the kitchen across all dynamic motions. In addition, confusion arises because the distribution of jumping data in the kitchen is now closer to the walking data than jumping

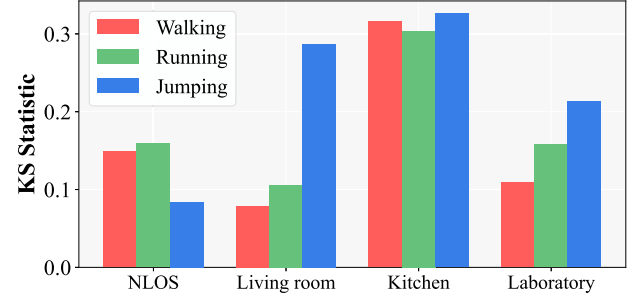


Fig. 6. Measurement of distribution discrepancies across domains with KS statistic (using bedroom data as reference).

TABLE VI
DG PERFORMANCE COMPARISON BASED ON TRAINING SETS ACROSS TEMPORAL CAMPAIGNS

Methods	AR-1d	AR-2a	AR-3a	AR-4a	AR-5a	AR-7a	Mean
AFFAR	0.9204	0.7998	0.6234	0.4866	0.8740	0.5743	0.7131
SHARP	0.8958	0.6772	0.5929	0.7031	0.8190	0.6598	0.7246
WiSR	0.8450	0.6085	0.6616	0.5781	0.7662	0.6423	0.6836
Meta-EMCAN	0.8889	0.6736	0.7150	0.5408	0.8514	0.5770	0.7078
Wi-ViTAL	0.8780	<u>0.7080</u>	0.8050	0.8315	0.9241	0.5500	0.7828

TABLE VII
DG PERFORMANCE COMPARISON BASED ON TRAINING SETS ACROSS DIFFERENT ENVIRONMENTS

Methods	AR-1b	AR-1c	AR-1d	AR-2a	AR-4a	AR-7a	Mean
AFFAR	0.5237	0.8523	<u>0.8622</u>	0.6593	0.6952	0.5560	0.6914
SHARP	0.3902	0.8169	0.8428	0.5838	0.6008	0.4898	0.6207
WiSR	<u>0.5807</u>	0.7183	0.8265	0.7008	0.4771	0.6598	0.6605
Meta-EMCAN	0.4834	0.8485	0.8213	0.7149	0.5913	0.7075	0.6945
Wi-ViTAL	0.6900	<u>0.8500</u>	0.8671	0.7710	0.5612	0.5876	0.7212

TABLE VIII
DG PERFORMANCE COMPARISON BASED ON TRAINING SETS ACROSS DIFFERENT ENVIRONMENTS AND PARTICIPANTS

Methods	AR-1b	AR-1c	AR-1d	AR-2a	AR-3a	AR-5a	Mean
AFFAR	0.5404	0.8433	0.9132	0.8274	0.8652	0.8265	0.8027
SHARP	0.5827	0.7440	0.8671	0.6765	0.6580	0.8203	0.7248
WiSR	0.3701	0.7611	0.8697	0.7643	0.5474	0.6597	0.6620
Meta-EMCAN	0.4042	0.6931	<u>0.9017</u>	<u>0.8099</u>	0.8321	0.8150	0.7427
Wi-ViTAL	0.5231	0.8467	0.8309	0.7352	0.9033	0.8467	<u>0.7810</u>

data in the bedroom, with KS statistic values of 0.10 and 0.33 respectively. This severe OOD shift explains why the kitchen experiences lower accuracy in activity classification. Despite this challenge, Wi-ViTAL still performs better in this situation than the other approaches.

Tables VI, VII, and VIII present the eight-label classification results across different scenarios. Table VI presents the DG performance using AR-1a, AR-1b, and AR-1c as input sets, highlighting temporal variations among source domains. Table VII extends the analysis by using AR-1a, AR-3a, and AR-5a as input sets, in which differences among source domains arise from both temporal and environmental aspects. Table VIII further expands

TABLE IX
COMPARISON OF PREDICTION ACCURACY AND COMPUTATIONAL COST ACROSS DIFFERENT MODELS

Methods	Metric	Est. inference time (GPU) [ms]	FLOPs	No. of model parameters	Memory usage [MB]	Five-label average accuracy [%]	Eight-label average accuracy [%]
AFFAR		1.69 ± 0.10	2.33×10^9	7.45×10^5	2.72	94.65	73.57
SHARP		1.29 ± 0.06	1.62×10^8	1.28×10^5	0.49	95.37	69.00
WiSR		1.87 ± 0.09	1.06×10^9	3.74×10^6	14.39	92.11	66.87
Meta-EMCAN		7.45 ± 0.19	6.58×10^8	2.71×10^5	1.07	93.53	71.50
Wi-ViTAL		2.10 ± 0.09	6.07×10^8	5.12×10^6	19.68	97.57	76.17

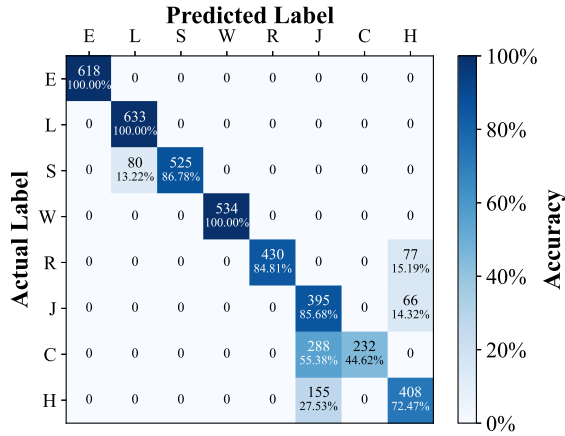


Fig. 7. Representative confusion matrix for the eight-label classification task.

the analysis to include temporal, environmental, and participant variation, with AR-1a, AR-4a, and AR-7a as input sets. Since the models perform effectively on the training data, the 20% test split from source domains for testing naturally yields high accuracy. Therefore, to properly evaluate DG performance in the eight-label classification task, the evaluation focuses on testing in unseen domains.

The eight-label classification introduces additional activities, including standing still, sitting down/standing up, and arm exercises, which further increase the complexity of the task. Although the preprocessing focuses on extracting domain-invariant features, activities such as empty, sitting, and standing, which share similar static Doppler patterns, are easily confused with one another. Misclassifications also occur in some unseen domains, where jumping, sitting down/standing up, and arm exercises resemble each other due to arm swing, yielding similar velocity signatures in the Doppler traces. In addition, severe OOD shifts make it difficult for the model to generalize effectively in the eight-label classification. Fig. 7 shows a representative result for eight-label recognition (e.g., AR-1c in Table VII), clearly highlighting the misclassifications discussed above. Despite these challenges, Wi-ViTAL generally offers higher or comparable accuracy than other approaches. Because AFFAR is designed for user-independent DG, it achieves better overall performance when the input sets include participant variation, with an accuracy about 2% higher than that of Wi-ViTAL in Table VIII. Moreover, Wi-ViTAL performs better on five-label and eight-label classification tasks compared with WiSR and Meta-EMCAN, both of which employ GRL as the adversarial learning module for DG. It can support the development of smart building applications, including assisted living in private homes

and intelligent monitoring systems for laboratories and offices. Although Wi-ViTAL has been evaluated in several representative indoor settings, assessing its performance in a wider range of real-world scenarios remains essential.

E. Evaluation of Computational Cost and Prediction Accuracy

Table IX compares the computational cost and prediction accuracy of five different models, under the experimental setup described in Section V-B. AFFAR, SHARP, and Meta-EMCAN only require a small amount of memory (2.72, 0.49, and 1.07 MB, respectively) and are suitable for deployment on edge devices. Although the proposed Wi-ViTAL has the largest number of parameters, it requires only 19.68 MB of memory. Notably, only 1.46% of its parameters belong to the ViT components since the multi-scale $Q/K/V$ tokens are generated using DWConv and GConv. Moreover, the FLOPs of Wi-ViTAL are lower than those of AFFAR, WiSR, and Meta-EMCAN, indicating that, despite having more parameters, its computational cost remains relatively low due to the efficient convolutional operations and linear activation functions. As a result, Wi-ViTAL achieves a competitive inference time, averaging 2.1 milliseconds per sample on an NVIDIA RTX 3090 GPU, which makes it a promising candidate for real-time recognition. This computational cost is based on the five-label setting, as the eight-label setting affects only the final fully connected layer, resulting in minimal differences. More importantly, Wi-ViTAL outperforms them in classification accuracy, particularly in domains that differ more from the source domains, which makes Wi-ViTAL more practical for real-world applications. Furthermore, the memory usage (19.68MB) and computational cost (6.07×10^8 FLOPs) of Wi-ViTAL are acceptable, as this model size is easily manageable for typical edge devices. Overall, Wi-ViTAL offers a better trade-off between accuracy and computational cost for WHAR.

F. Ablation Study

Fig. 8 presents the ablation study for both five-label and eight-label classification tasks, evaluating Wi-ViTAL's performance on target domains. This study examines the effects of different module combinations (C_*), including the feature extractor (G_θ), activity classifier (C_φ), domain classifier (D_ψ), as well as enhancements via AdaIN and adversarial learning with MDA and CDA. The training sets used are AR-1a, AR-1b, and AR-1c.

In the five-label classification task, incorporating the domain classifier (D_ψ) into the initial model (G_θ & C_φ) has a minimal effect on activity classification results. However, after integrating AdaIN to bias features toward domain information prior to the domain classifier, the model can extract domain-specific

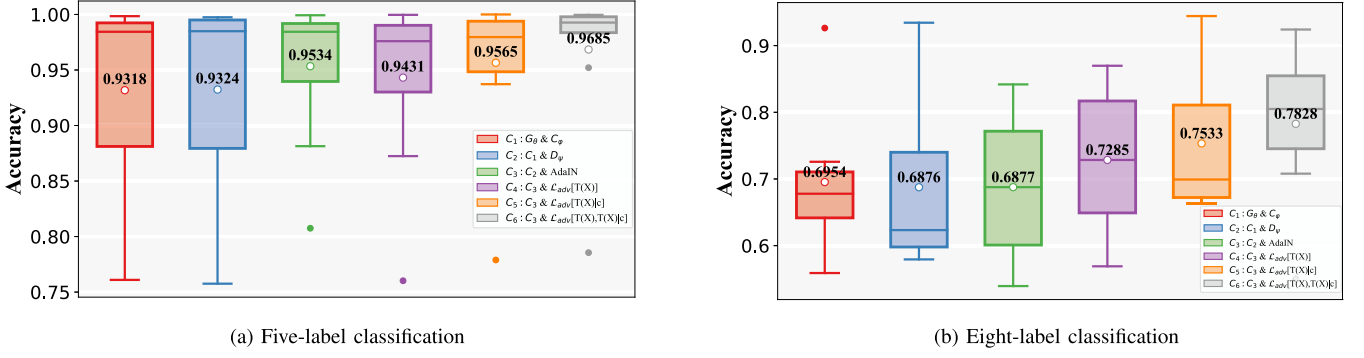


Fig. 8. Impact of feature extractor with separate and combined use of activity classifier, domain classifier, AdaIN, and adversarial learning modules (MDA and CDA).

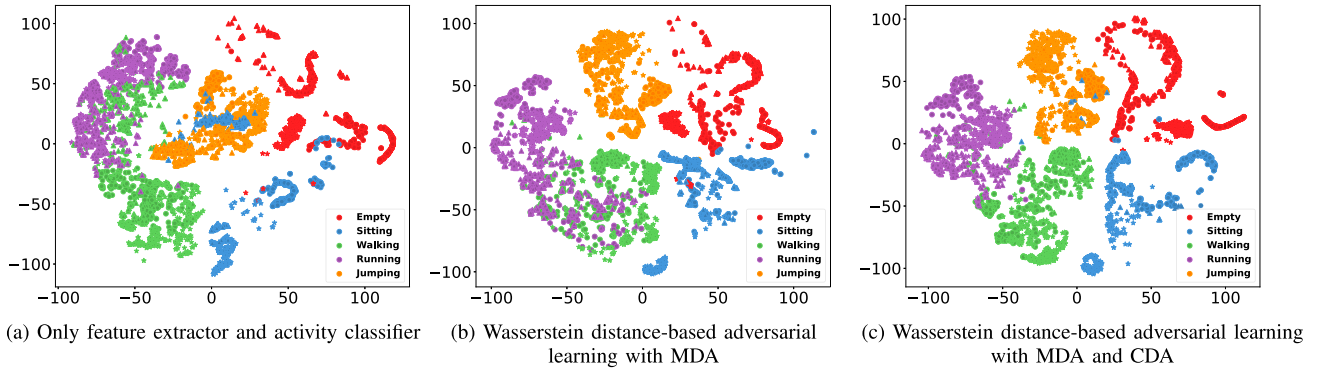


Fig. 9. Visualization of t-SNE embeddings for unseen test sets in bedroom, living room, and lab; classes represented by colors and domains by shapes (bedroom: o, living room: *, lab:).

information while still capturing behavioural patterns, which enhances its generalization across different domains. It is also observed that considering MDA or CDA individually does not substantially improve classification performance, but combining both MDA and CDA alongside Wasserstein distance notably enhances the model's generalization, achieving an average accuracy of 96.85% across AR-1d to AR-7a. After applying adversarial learning with MDA and CDA, the grey box displays a narrow IQR of just 1.43% , with the lower whisker extending to 0.05% . In contrast, the red box presents a considerably wider IQR of 11.13% , with its lower whisker extending to 12.02% . For eight-label classification, simply combining the domain classifier with AdaIN does not improve the average classification accuracy. However, under the C_6 configuration, behavioural patterns can also be mapped to a shared latent space, strengthening the model's ability to predict human activities in target domains. Consequently, the grey box demonstrates higher accuracy and lower variability in both the five-label and eight-label classification tasks. Its narrow IQR and shorter whiskers indicate more consistent and stable performance with fewer fluctuations.

Fig. 9 shows the t-SNE visualizations of dimension-reduced features, where the features correspond to the logits of the activity classifier before softmax and Cross-Entropy calculations. In Fig. 9, the activity clusters, e.g., in the five-label classification task, are more clearly separated in Wi-ViTAL's outputs, compared to scenarios where adversarial learning is

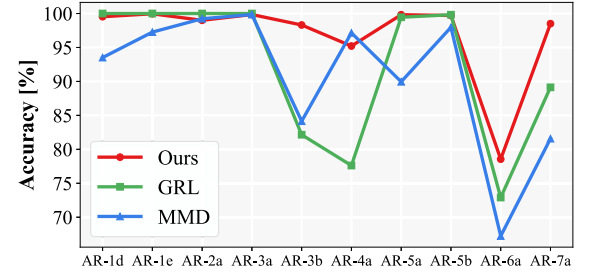


Fig. 10. Performance comparison of GRL, MMD, and our Wasserstein distance-based approach in adversarial learning modules for five-label classification across test domains.

absent or only the marginal distribution is aligned. Furthermore, the domain boundaries within the clusters are blurred, indicating that the extracted behavioural features are generalized across different domains.

Fig. 10 displays the five-label classification accuracy across different domains, obtained under the same model architecture as Wi-ViTAL and the training scenarios outlined in Table IV, with the proposed Wasserstein distance-based adversarial approach replaced by GRL in [13] or MMD in [37]. As illustrated, both GRL and MMD exhibit poorer generalization performance compared with the proposed approach, particularly when applied to unseen target domains that differ significantly from the training sets, such as NLOS scenarios, kitchen, and laboratory.

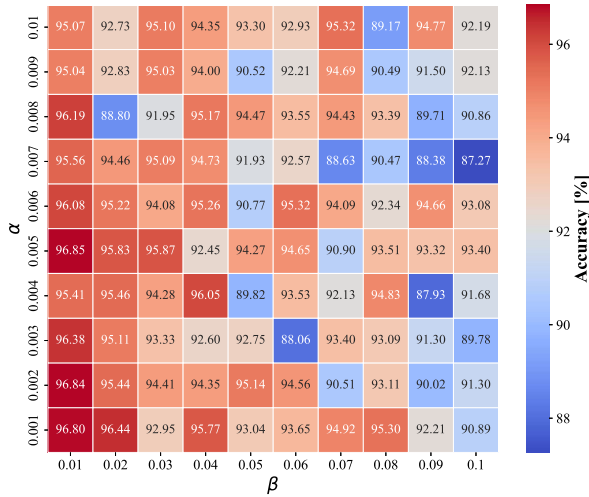


Fig. 11. Average accuracy of five-label classification in unseen domains with different hyperparameter combinations for MDA and CDA.

The proposed Wasserstein distance-based adversarial module provides more stable gradients than both GRL and MMD. GRL is prone to the gradient vanishing issue, as the domain classifier's efforts to improve its accuracy during training can lead to trivial gradients, causing the adversarial learning to become ineffective. Additionally, GRL simply reverses the sign of the gradients; however, this does not inherently enable the feature extractor to identify generalized patterns, since treating source domains as target domains and vice versa will also increase the loss of the domain classifier and decrease the total loss by adding its negative. As a result, it merely separates domains instead of merging their feature distributions into a domain-invariant representation. MMD, on the other hand, relies on kernel selection and mean embeddings of distributions, which may fail to capture domain discrepancies that involve higher-order characteristics such as variance, skewness, and kurtosis [50].

G. Sensitivity Analysis of Adversarial Learning Parameters

The selection of α and β in (12) affects the performance of adversarial learning between the feature extractor and the domain classifier, further impacting the model's ability to recognize behavioural patterns, as shown in Fig. 11. The hyperparameter α for MDA is analyzed over the range of 0.001 to 0.01, while the hyperparameter β for CDA is examined between 0.01 and 0.1. Each value in the heatmap indicates the average five-label classification accuracy across the target domains, specifically for sets ranging from AR-1d to AR-7a.

As observed, the model's performance is fairly sensitive to the selection of α and β , with some fluctuations in accuracy as these parameters change. The highest accuracy, 96.85%, is achieved when α and β are equal to 0.005 and 0.01. These two hyperparameters are used consistently throughout the experiments. As both α and β increase, the model's accuracy tends to decline. Higher classification performance is concentrated in the lower-left region of the heatmap. Notably, when $\beta = 0.01$, the average accuracy remains above 95%, regardless of the value of α .

VI. CONCLUSIONS AND FUTURE WORK

In this paper, Wi-ViTAL, a lightweight ViT model incorporating a multi-scale ReLU linear attention mechanism and Wasserstein distance-based adversarial learning module, is proposed to improve DG performance. The experiments involving domain diversity across different days, environments, participants, and obstacle configurations have demonstrated that the proposed Wi-ViTAL model achieves 97.57% average accuracy for five-label classification and more than 76% for eight-label classification in unseen domains. An overall improvement in accuracy has also been observed when compared with the benchmark models. The ablation study also reveals that Wi-ViTAL reduces the IQR from 11.13% to 1.43% in five-label classification, while achieving a significant overall performance improvement in eight-label classification when all proposed blocks are used. Based on the sensitivity analysis, the hyperparameter of CDA is set to 0.01, which is identified as the most suitable for WHAR, achieving the best performance when combined with an MDA hyperparameter of 0.005. These results provide valuable insights for both current and future deployments of passive, device-free wireless sensing systems, particularly for WHAR applications.

However, Wi-ViTAL still faces challenges in distinguishing activities such as jumping, sitting down/standing up, and arm exercises in unseen domains due to severe OOD shifts. Future research aims to improve the model's generalization ability by exploring additional domain-invariant characteristics beyond Doppler traces in the CSI preprocessing stage, enabling better differentiation of complex activities across multiple domains. In addition, recognizing unknown activities outside the target categories in unseen domains is crucial, as the vast diversity of human activities makes it impractical to include all possible actions. Moreover, deploying the proposed offline model to edge devices for real-time recognition in real-world scenarios remains a challenge to be addressed.

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